



Demolition of Barriers Posed by Social Coverage and Financial Sustainability: An Evaluation of the Health Insurance System in Algeria



Khayreddine Bouzerb^{1*}, Hassiba Almi², Aboubaker Khoualed², Asma Derradji³

¹ Department of Economic Sciences, Faculty of Economic, Commercial and Management Sciences, Mohamed Seddik Benyahia-Jijel University, 18000 Jijel, Algeria

² Department of Economic Sciences, Faculty of Economic, Commercial and Management Sciences, Badji Mokhtar-Annaba University, 23000 Annaba, Algeria

³ Department of Economic Sciences, Faculty of Economic, Commercial and Management Sciences, Chadli Bendjedid-El Tarf University, 36000 El-Tarf, Algeria

* Correspondence: Khayreddine Bouzerb (k.bouzerb@univ-jijel.dz)

Received: 04-05-2026

Revised: 05-21-2026

Accepted: 06-03-2026

Citation: Bouzerb, K., Almi, H., Khoualed, A., & Derradji, A. (2026). Demolition of barriers posed by social coverage and financial sustainability: An evaluation of the health insurance system in Algeria. *J. Corp. Gov. Insur. Risk Manag.*, 13(2), 69–81. <https://doi.org/10.56578/jcgirm130201>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: The present paper aims to evaluate Algeria's health insurance system in terms of its mandatory and supplementary components, given its pivotal role in achieving universal health coverage and upholding the principle of social solidarity. To pursue their key objectives, the scholars adopted an analytical approach to assess the existing state of these two health insurance systems and analyse their respective performances. The significant gains from mandatory insurance were highlighted, particularly in the expansion of demographic coverage. Besides, supplemental insurance played a supportive role in covering the remaining costs borne by the insured and in alleviating their pressing financial burden. The findings from the analysis revealed bottlenecks that hindered the sustainability of the system, ranging from ongoing financial pressure exerted by the funds due to escalating healthcare costs, the prevalence of chronic diseases, as well as the notable gap between approved reimbursement reference rates and the actual costs incurred by various treatments. These unprecedented challenges necessitate immediate action to develop novel alternative financing mechanisms for augmenting the funds' resources, to update the reimbursement reference rates, and to accelerate digitisation to rationalise medical expenditures and strengthen oversight.

Keywords: Social security; Mandatory insurance; Health insurance system; Reimbursement reference rates

1. Introduction

Demographic growth and imposing health challenges in the previous decades have prompted governments to reexamine the relationship between economic development and social protection, particularly in view of rising healthcare costs, increasing rates of chronic diseases, and growing demands for quality life. The predicament created by these multiple problems are no longer a social issue; they carry notable economic and financial implications, thus urging governments and relevant institutions to consider mechanisms for directing resources toward more efficient and sustainable health care systems. Such an operation requires mobilising massive investments and constructing innovative financing mechanisms. Recent literature has confirmed that health insurance is a key mechanism for supporting this trend by providing comprehensive coverage and mitigating the financial risks individuals face, thereby contributing to the achievement of sustainable development goals and ensuring the stability of human capital.

In this context, Algeria's experience stands out in social protection, provided that its early adoption of laws and policies enshrined the principles of national solidarity and social justice, and the establishment of numerous legislative and regulatory frameworks governed the insurance system. These development aim to expand demographic coverage and improve the quality and scope of benefits, while accommodating various groups

through specialised funds, reflecting a growing trend toward integrating all segments of society into the social protection network.

The Algerian experience is not limited to the legislative aspect alone; it also extends to developing tools and mechanisms designed to facilitate access to care and modernise services. Among the most notable of these tools are the implementation of supplementary health insurance (mutual aid societies), mechanisms for contracting with the private medical sector, as well as the third-party payment system via the “Chifa Card”. The emergence of these tools is imperative because it reflects a shift in protection policy from merely providing coverage to using digital transformation as a tool to control expenditures, improve the quality of care, and safeguard the financial stability of the funds.

The Algerian experience as a system should, inevitably, seek to reconcile social gains with pressing financing challenges in light of current demographic and epidemiological shifts. Consequently, the research question arose in this study: **To what extent could the health insurance system in Algeria utilize mandatory and supplemental coverage mechanisms to achieve universal health coverage for members of society?** This question was further examined via these three sub-questions. First, what is the impact of mandatory health insurance in Algeria, and to what extent is it effective in providing health coverage? Second, what is the reality of supplemental health insurance in Algeria, and what is its role in enhancing health insurance coverage? Third, what are the requirements for developing health insurance systems in Algeria to provide greater social coverage on the one hand and enhance the financial sustainability of the funds on the other?

This study aims to analyse the Algerian experience by examining the legislative framework related to social security, assessing the current state of both mandatory and supplemental health insurance, reviewing key performance indicators (such as the growth in the number of insured individuals), and then analysing the challenges encountered by this system, while emphasizing the development requirements that could contribute to rationalizing expenditures and ensuring the financial sustainability of the funds. The contribution of this study lies in providing a comprehensive assessment of the combined mandatory and supplemental insurance, rather than relying on previous partial analyses that explicitly distinguished between mandatory social security and supplemental insurance.

2. The Concept and Forms of Health Insurance Systems

Health insurance is a primary enabler of access to health as it provides financial protection for underprivileged people, promotes access to health care services, connects to improved health outcomes and well-being as well as reducing out-of-pocket expenses, especially amid elevating treatment costs. The World Health Organisation (WHO) estimated that 150 million people worldwide suffered catastrophic financial crises annually due to out-of-pocket expenses incurred to meet their imminent health care needs (Ho, 2015; WHO, 2005). The concept of health insurance first emerged in 1694, pioneered by Hugh Chamberlen the Elder, a descendant of Peter Chamberlain. In the 19th century, the emergence of accident insurance operated on a mechanism very much similar to modern disability insurance. This payment model persisted until the early 20th century. From the mid- to late 20th century, traditional disability insurance evolved into modern health insurance programs. Today, the most comprehensive health insurance programs cover routine, preventive, and emergency healthcare costs (Kapur, 2024).

Health insurance is an economic mechanism designed to compensate for imperfect markets in the healthcare sector; it is based on the principle of transferring risks arising from the uncertainty of illness from an individual to a broader risk pool (Arrow, 1978). Moreover, health insurance serves as a social protection system that pools revenues and shares financial risks, ensuring the access to medical care without confronting financial hardship (Kutzin, 2013). Health insurance is crucial in numerous ways but its greatest value lies in protecting against financial shocks; without insurance, families must pay for treatment out of their own pockets, leading to substantial health expenditures (Cutler & Zeckhauser, 2000). Therefore, health insurance systems play a vital role in redistributing wealth and risk between the healthy and the sick, removing financial barriers to obtaining affordable medical services, and ameliorating public health indicators; they in turn facilitate human capital accumulation and economic productivity (Bloom et al., 2004).

Health insurance systems vary across countries in financing mechanisms, the degree of mandatory coverage, and the nature of the governing body. These systems embody the discrete features set out below:

(i) Social health insurance: Historically known as the Bismarck model, it is the cornerstone of social protection in many countries. This system is mandatory and is typically linked to formal employment. It is primarily funded through contributions or payroll deductions, with employers and employees sharing costs in proportions determined by law. The coverage under this system is based on the principle of social solidarity, whereby financial contributions represent a percentage of income (based on the ability to pay). Meanwhile, access to health care is linked to medical need, regardless of the amount of contribution (Saltman et al., 2004).

(ii) National health Insurance: Recognized as the Beveridge model, it indicates direct state intervention in the health sector. The state fully funds this system through tax revenues, rather than direct individual contributions. It provides comprehensive health coverage to all citizens as a constitutional and fundamental right, without

associating coverage with employment status (Mossialos et al., 2002).

(iii) Private health insurance: Operates in compliance with the intense market mechanisms and differs radically from social and national systems; it is defined as a voluntary insurance arrangement administered by commercial insurance companies or mutual insurance societies (Sekhri & Savedoff, 2006).

3. Methodology

According to Manjunatha (2019), the descriptive method could appropriately tackle the occurrence of various social phenomena, especially those that are economic, administrative, or financial in nature, to achieve the most precise outcomes.

This research employed a triangulation approach to assess the performance, coverage dynamics, and financial sustainability of the health insurance system in Algeria, drawing on three primary sources listed below:

(i) Official statistical and institutional reports: These provided on-the-ground indicators, demographic data, and financial statements issued by the relevant national agencies. They were the National Social Insurance Fund for Salaried Workers (CNAS), the National Social Security Fund for Non-Salaried Workers (CASNOS), and the National Insurance Council (CNA).

(ii) The legislative and regulatory framework: A thorough review was conducted on the legal texts governing mandatory and supplementary insurance, as well as the implementing decrees specifying contribution rates and reference rates for reimbursement of medications and health services.

(ii) Economic Literature and Studies: These included peer-reviewed academic research in health economics as well as reports from specialized international and regional bodies such as the International Social Security Association (ISSA) and the Institute for Economic Forecasting of the Mediterranean (IPEMED).

The time frame demarcated in this study covered the period from 2011 to 2025, to allow tracking of reforms and structural changes in Algeria's health insurance system, both mandatory and supplementary, enlightened by the "aggressive" demographic and health-related developments.

4. The Regulatory and Legislative Framework of the Health Insurance System in Algeria

Algeria's health risk coverage is based on a dual system composing of mandatory social insurance and voluntary (private) supplemental insurance. The international welfare models were closely aligned with the country's economic and social reality. This system becomes the source of financial protection against illness, maternity, disability, and occupational risks, while promoting social justice and solidarity (Boukhedimi et al., 2025).

More importantly, Algeria's health insurance system works upon laws, orders, and decrees that define coverage mechanisms and conditions while regulating reimbursement methods, thus reflecting public policy orientations and the state's efforts to maintain social protection for members of society. The regulatory framework for compulsory health insurance and supplementary insurance in Algeria is depicted below.

4.1. The Legal Framework Governing Mandatory Health Insurance in Algeria

Algeria's compulsory health insurance system relies on an all-encompassing legal and regulatory framework, founded primarily in the early 1980s and amended multiple times to keep pace with economic and demographic developments. According to the Ministry of Labour, Employment and Social Security (MTESS), It aims to guarantee sustainable financing and expand coverage based on the governing laws (MTESS, n.d.):

- Law No. 83-11 dated 21 Ramadan 1403 AH, corresponding to July 2, 1983, on social insurance, which established the general framework for the national social protection system in Algeria.
- Law No. 83-13 dated 21 Ramadan 1403 AH, corresponding to July 2, 1983, concerning work-related accidents and occupational diseases, which aims to compensate workers for injuries sustained while performing their duties.
- Law No. 83-14, dated 21 Ramadan 1403 AH, corresponding to July 2, 1983, concerning the obligations of parties subject to social security, which defined the duties of employers and insured persons toward social security funds.
- Law No. 83-15, dated 21 Ramadan 1403 AH, corresponding to July 2, 1983, concerning social security disputes, which regulates the methods for resolving medical and administrative disputes that may arise among social security agencies, insured persons, and employers.

4.2 The Legal Framework Governing Supplementary Health Insurance in Algeria

Supplementary health insurance in Algeria is based on two fundamental laws, one of which regulates mutual insurance societies whereas the other regulates insurance companies:

- Decree No. 95-07 on Insurance, as amended and supplemented by Law No. 06-04, which regulates the

insurance market and governs personal insurance contracts and supplemental insurance intended to address gaps not covered by insurance funds (Mabrouk, 2019).

- Law No. 15-02 of February 15, 2015, on social mutual aid societies (which repealed the previous Law No. 99-06). This law regulates the operations of funds and mutual aid societies about supplemental and voluntary health insurance, whereby mutual aid societies absorb the remaining financial shortfall not compensated by the primary fund (CNAS/CASNOS) at a rate of up to 20%.

5. Origins of the Health Insurance System in Algeria and the Groups Covered

As the most crucial component of social security in Algeria, the health insurance system offers coverage for risks associated with illnesses or injuries that necessitate health care. Initially, this system benefitted only the economically active population; however, expansion since the 1983 laws has put several specific groups under the list. In collaboration, students, people with disabilities, professionals, etc. have transformed it from an occupational system into a social system (Ziani, 2020).

The evolution of the Algerian health insurance system could be categorized into three key stages since its inception (Sifer & Guehairia, 2024). These developmental phases are outlined in the Table 1.

Table 1. Stages of development undergone by the health insurance system in Algeria

Period	Developments of the Health Insurance System
Establishment and post-independence period (1962–1973)	• The introduction of AMG and the establishment of insurance funds covering the risks of illness, maternity, and disability, with limited and restricted coverage. • After independence, the National Social Security Fund was established, merging 11 pre-existing insurance schemes. The free medical assistance system continued to operate, placing a substantial financial burden on the state budget. • Between 1974 and 1982, the Social Security Fund became directly involved in financing the healthcare sector through the management and monitoring of the hospital forfait system.
Reforms and unification of the system (1974–1991)	• In 1983, the social security system was unified to eliminate administrative disparities. Nevertheless, health insurance and social security budgets faced inconceivable financial pressure due to rising costs of healthcare provided abroad. • The establishment of specialised funds, such as CASNOS (1992) for self-employed workers, while CNAS focused on providing health and social coverage for salaried employees.
Diversification of funding sources and autonomy of the funds (1992–present)	• Starting from 2010, new funding sources were introduced for the National Social Security Fund, linking healthcare financing to public health through taxes on tobacco products and a 5% levy on the net profits of imported medicines.

Note: AMG = Free Medical Assistance; CASNOS = National Social Insurance Fund for Non-Salaried Workers; CNAS = National Social Insurance Fund for Salaried Workers.

Several contributing factors are propelling the development of health insurance in Algeria, as listed below (Statista, n.d.):

- **Customer Preferences:** Customers in Algeria are increasingly opting for health insurance plans that go beyond coverage for hospitalisation and medications to include additional benefits, such as dental care, maternity services, and preventive medical checkups.
- **Market Trends:** The rise of private health insurance companies is one of the most prominent trends in the Algerian market, as they offer more flexible and personalised health insurance options tailored to individual preferences and lifestyles. Besides, the sector is progressing toward digitalisation, with more interested companies participate in online services round-the-clock for purchasing insurance policies, filing claims, and accessing medical and health resources.
- **Demographic Profile:** Algeria has a unique demographic profile, characterised by a high proportion of young people with a growing awareness of the importance of health and wellness. This shift is driving demand for health insurance products.
- **Macroeconomic Factors:** Several macroeconomic factors lead to the growth of the health insurance market in Algeria, including stable economic growth, rising disposable income, and the government's ongoing efforts to develop and modernise its healthcare infrastructure.

As a result, two main groups are the direct beneficiaries of health insurance (Bougrine, 2005):

- **Insured individuals:** salaried workers regardless of their sectors of activities; self-employed workers; former workers receiving social security benefits (old-age or retirement pensions, benefits for work-related accidents or occupational diseases, and unemployment insurance benefits), as well as certain individuals whose status grants them the status of insured persons (students, professionals, veterans, and the indigent receiving state social assistance).

- Health insurance beneficiaries with dependent status: spouses, dependent children (up to age 18 in all cases, age 21 for students, and age 25 for professionals, with no age limit for people with disabilities and daughters living at home), as well as dependent parents when their income does not exceed the minimum retirement pension amount.

According to the National Economic, Social and Environmental Council (CNESE), the expansion of social protection has extended health insurance coverage to previously uncovered occupational and population groups, including members of the Algerian diaspora, artists, actors, seafarers, and fishers (CNESE, 2024). Furthermore, on July 20, 2025, the Ministry of National Solidarity, Family, and Women's Affairs announced that, effective July 27, 2025, indigent individuals and vulnerable groups not covered by social security could apply for free medication under Executive Decree No. 24-287 dated August 2024 (United Nations Economic & Social Commission for Western Asia, 2025). These initiatives have allowed a significant segment of the population access to health care and protection against risks associated with health and ageing, covering more than one-third of members of the National Social Insurance Fund for Wage Earners and equivalent to 37% of the state-covered contributions. This measure helps eradicate health disparities and advocate social well-being nationwide (CNESE, 2024).

6. Mandatory Health Insurance in Algeria

The MTESS administers Algeria's mandatory health insurance system through two main executive bodies: the CNAS and the CASNOS.

6.1 National Social Insurance Fund for Salaried Workers

The National Social Insurance Fund for Salaried Workers is a public institution with private-sector management, in accordance with Article 49 of Law No. 88-01 dated January 12, 1988, which contains the framework law for public economic institutions. It represents legal personality and financial autonomy. The National Social Insurance Fund administers social security for salaried workers in Algeria and ensures the provision of benefits related to sickness, maternity, work-related accidents and disability, as well as various forms of social assistance for insured persons and their beneficiaries (CNAS, 2018).

The CNAS' health expenditures could be further divided into the following components in calculation:

- Daily Benefits: These are benefits related to time off work, which the CNAS pays.
- Referrals for Treatment Abroad: Care provided abroad relates to intractable diseases that entail high costs. Public authorities have adopted measures to minimize these expenditures by limiting the number of diseases that qualify for referral abroad, diversifying the host countries, and motivating foreign medical teams to come to Algeria.
- Hospitalisation Flat Rate: Since 1947, under the Law on Free Medical Care in Algeria, social security has been required to contribute to the operating budgets of healthcare institutions through a flat-rate contribution set annually under the Finance Act (Ziani, 2020).
- Medication Reimbursements: This is the largest expenditure item for the CNAS, as the health card allows the insured person and their dependents to benefit from a third-party payment system for any prescription, provided its value does not exceed 3,000 Algerian dinars (DZD) or for the first two prescriptions within three months. For prescriptions exceeding this limit, the cost must be paid upfront, and a reimbursement claim must then be filed with the Social Security Fund (APRIL International, n.d.)

Except for chronic illnesses, which are covered and reimbursed at 100%, other health services are reimbursed at 80% of the reference rate approved by the Algerian Social Security system. The insured person must pay the costs upfront, unless the doctor or healthcare facility is contracted under the third-party payment system (direct payment) (APRIL International, n.d.)

The financing of the fund depends predominantly on the employer's and employee's contributions operating within a pay-as-you-go system, a mechanism that continues to serve as the foundational pillar for social insurance frameworks globally. According to Executive Decree No. 15-236 of September 3, the statutory contribution rates apportioned among employers, employees, and social services were established and detailed in the Table 2.

The table shows that, although employees bear a portion of the deductions, employers bear the lion's share; furthermore, in some sectors, employers fund contributions exclusively.

In addition to contributions from workers and employers, public authorities play a crucial role in ensuring the system's sustainability through supplementary funding sources, which include the following (Hamadi & Abrika, 2023):

- Government subsidies: The Algerian government allocates a budget to fund social security, and this budget is specifically earmarked to finance certain services and benefits, such as family allowances and expenditures known as "national solidarity expenditures".
- Other sources of funding: These include returns on invested funds, contributions paid by beneficiaries to establish their entitlements in the form of unemployment insurance and early retirement, surcharges, late

payment penalties, and other financial sanctions against delinquent employers, as well as donations and bequests.

Table 2. Employee's and employer's contributions to the CNAS

Category	Employer's Contribution	Employee's Contribution	Social Services Contribution	Total
Social insurance	11.5%	1.5%	–	13%
Occupational accidents and diseases	1.25%	–	–	1.25%
Retirement	11%	6.75%	0.5%	18.25%
Early retirement	0.25%	0.25%	–	0.5%
Unemployment insurance	1%	0.5%	–	1.5%
Total	25%	9%	0.5%	34.5%

Note: CNAS = National Social Insurance Fund for Salaried Workers. The en dash (–) indicates not applicable.

6.2 National Social Security Fund for the Self-Employed

A system for the self-employed has existed in Algeria since 1958; initially and until 1974, it was a pension-only system (Amrani, 2012). The fund was established by Executive Decree No. 92-08 dated January 4, 1992, as part of the institutional restructuring of the social security sector, to separate occupational categories and designate an independent body to manage the social risks specific to self-employed workers (ISSA, 2024). The Fund is exclusively responsible for providing social coverage to self-employed workers who carry out professional activities on their own account. This category includes self-employed professionals (such as doctors and lawyers), merchants, artisans, farmers, and partners in commercial firms. Enrolment in the Fund is mandatory by law once these involved parties commence their professional activities (Centre for European & International Social Security Liaison, 2024).

In health insurance, the CASNOS provides coverage limited to in-kind benefits for the insured person and their dependents (spouses and dependent children); the Fund does not provide cash benefits (such as daily sick leave compensation). According to Article 4 of Decree No. 96-17 dated July 6, 1996, amending and supplementing Law No. 83-11 dated July 2, 1983 on social insurance, in-kind benefits cover the costs of medical services, surgical procedures, medications, hospital care, laboratory and radiological examinations, dental care and prosthetics, prescription eyeglasses, spa treatments or specialized therapies related to specific illnesses, medical devices and prosthetics, functional rehabilitation, vocational rehabilitation, and ambulance transportation.

Additionally, the system covers 100% of the medical, pharmaceutical, and hospitalisation expenses related to prenatal care and childbirth for the insured woman or the insured person's spouse (MTESS, n.d.). It does not, however, provide financial compensation for time off work (cash benefits) as is the case for salaried workers.

Self-employed individuals are subject to a special contribution, with the total contribution rate borne by the taxpayer amounting to 15%. This is calculated based on annual taxable income or, in the absence thereof, on gross revenue or in certain cases, on the annual national minimum guaranteed wage. This percentage is divided equally (of 7.5%) between the social insurance and pension branches, and annual taxable income or gross revenue ranges from 216,000.00 DZD to 4,320,000.00 DZD. Annual contributions range from a minimum of 32,400.00 DZD to a maximum of 648,000.00 DZD (Hamadi & Abrika, 2023).

The differences of the health insurance system found between the self-employed workers and salaried employees are illustrated in Table 3.

6.3 Evaluation of the Mandatory Health Insurance System

Algeria's National Social Security System covers all nine branches stipulated in International Labour Organisation Convention No. 102. The assessment of Algeria's compulsory health insurance system implies the investigation of indicators that reflect the system's success in providing social protection in the health sector and in identifying its weaknesses or structural imbalances.

6.3.1 Increasing trend in the number of socially insured persons

The trend in the number of socially insured individuals reflected the level of demographic coverage in Algeria's social security system. The latest report from the National Statistics Office (ONS) indicated that the number of socially insured individuals in Algeria had shown an upward trend in both the National Social Insurance Fund for Employees and the National Social Security Fund for the Self-Employed, with the number of insured employees rising considerably from 8,819,160 in 2011 to 12,316,693 in 2017, while the proportion of the economically active population declined slightly from 57.27% in 2011 to 47.69% in 2017. As regards the number of self-employed insured persons, it rose substantially from 1,776,581 in 2011 to 2,886,598 in 2017 (ONS, 2017).

Presenting data from the CNESE (2024), Table 4 delineates the annual progression of socially insured populations in Algeria between 2020 and 2023, disaggregated by fund affiliation (CNAS and CASNOS) and pensioners' status.

Table 3. Comparison of the health insurance system for the self-employed and salaried employees

Criterion	CNAS	CASNOS
Target group	Salaried employees	Self-employed workers and liberal professionals
Benefits	In-kind and cash benefits	In-kind benefits only
Financing	Contributions from employees and employers, together with government subsidies	Workers' contributions
Mandatory nature	Mandatory for all salaried employees	Mandatory for all registered non-salaried workers
Cash benefits	Sickness benefits (daily compensation for loss of income), maternity benefits (5 months), and disability pensions	No cash benefits for sickness, disability, or occupational accidents
In-kind benefits	Medical consultations, hospitalisation, surgery, pharmaceutical products, dental care, optical care, medical devices, and preventive care	Medical consultations, hospitalisation, surgery, pharmaceutical products, dental care, optical care, maternity care, and preventive care
Financing mechanism	Contributions shared between the employee and employer, with government subsidies	Fully dependent on contributions from self-employed workers, with limited government support

Note: CNAS = National Social Insurance Fund for Salaried Workers; CASNOS = National Social Security Fund for Non-Salaried Workers.

Table 4. Trends in the number of socially insured persons in Algeria for the period of 2020–2023

Year	CNAS Insured Persons	Pensioners Inclusive	CASNOS Insured Persons	Pensioners Inclusive
2020	11,618,605	2,909,821	2,099,909	324,428
2021	11,435,376	2,996,660	2,037,312	345,497
2022	13,994,360	2,860,359	1,931,122	358,840
2023	14,348,200	2,917,810	1,899,353	374,093

Note: CNAS = National Social Insurance Fund for Salaried Workers; CASNOS = National Social Security Fund for Non-Salaried Workers.

Table 4 highlights distinct trajectories regarding the evolution of socially insured individuals across the two schemes over the 2020–2023 period, refuting the assumption of synchronized and parallel growth across both funds. For the salaried scheme (CNAS), total insured persons (inclusive of pensioners) experienced a temporary contraction of 1.58% in 2021 (declining from 11,618,605 in 2020 to 11,435,376)—a consequence of COVID-19 pandemic disruptions and formal labor market shocks—before recording a sharp rebound of 22.38% in 2022 (13,994,360) to reach 14,348,200 in 2023. This expansion broadened the potential beneficiary base, driving health insurance expenditures upward through intensive pharmaceutical consumption via the Chifa Card system and rising demand for specialized medical services.

In contrast, the non-salaried scheme (CASNOS) exhibited a continuous decline; total insured individuals fell steadily from 2,099,909 in 2020 to 1,899,353 in 2023. Crucially, while the overall membership base of CASNOS eroded, the non-salaried pensioner cohort witnessed steady growth of 15.31% (rising from 324,428 to 374,093), which elevated the relative share of pensioners from 15.45% to 19.70% of total non-salaried insured individuals.

Across both systems, the total number of affiliated pensioners exceeded 3.29 million by 2023 (2.92 million in CNAS and 374,000 in CASNOS). This demographic segment represents a primary cost driver, as older age cohorts are disproportionately affected by chronic pathologies requiring long-term, 100%-reimbursed therapeutic coverage. Under the contribution-based pay-as-you-go system, the structural contraction of the active contributor base in CASNOS, coupled with the expanding volume of high-cost chronic care beneficiaries, exerts persistent actuarial pressures on the financial equilibrium of Algeria's social security system.

Figure 1 illustrates the share of socially insured groups at the end of the first quarter of 2023.

The figure indicates that the special and included categories encompassing retirees, accounted for the largest share of 57%, of which 21% were retirees, while the economically active population accounted for merely 43% in total, thus reflecting a heterogeneous composition that foreshadows future financial imbalances. Since the rising numbers of retirees and the elderly are closely linked to an increase in chronic and costly diseases such as heart disease, kidney failure, and cancer, the likelihood of providing more costly services would lead to a concentration of health spending on medications. These expenditures accounted for more than 90% of the funds' total health expenditures, which adversely impact the financial situation though the working-age population declines.

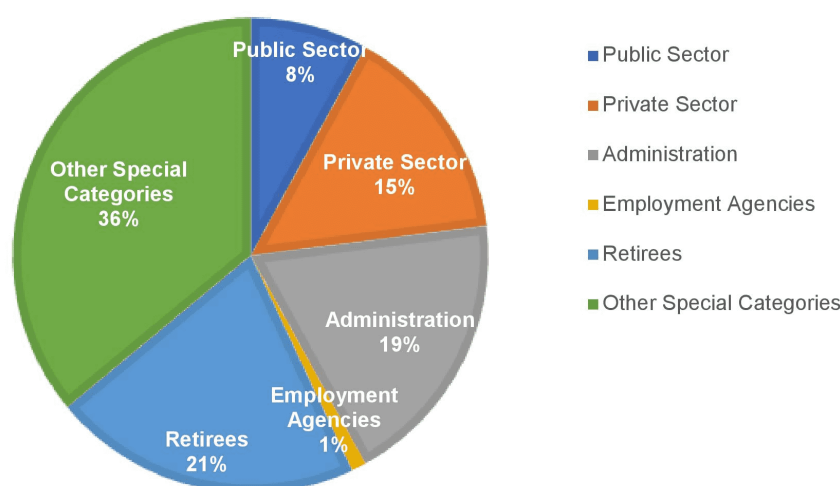


Figure 1. Percentage of socially insured persons by category (as of March 31, 2023)

6.3.2 Financial status of social security funds

According to the ONS (2017), the annual revenues, expenditures, and net financial balances recorded between 2011 and 2017 are shown in Table 5.

Table 5. Financial balance of the CNAS for the period of 2011–2017 (in millions of DZD)

Indicator	2011	2012	2013	2014	2015	2016	2017
Revenue	374,138	477,285	423,838	459,757	474,944	482,065	492,340
Expenditure	240,607	280,753	322,523	374,060	395,395	400,553	435,227
Balance	+133,531	+196,532	+107,315	+85,697	+79,549	+81,512	+57,113

Note: CNAS = National Social Insurance Fund for Salaried Workers; DZD = Algerian dinars.

The latest report from the ONS (2017) commented that the fund's financial position was positive as revenues exceeded expenditures, thus ensuring coverage for its members. However, the fund ran a deficit in subsequent years as expenditures rose amid the COVID-19 pandemic, resulting in a structural deficit of nearly 155 billion DZD before 2022. By 2024, the fund's revenues reached 1,827 billion DZD, when compared with 1,658 billion DZD in 2023, while direct health insurance expenditures amounted to 630 billion DZD in 2024 (Eco Times, 2025).

The revenues of the National Social Insurance Fund for Wage Earners were distributed as follows. The fund's share covered its expenses of 793 billion DZD, in support of the Pension Fund of 734 billion DZD); the supplementary contribution to finance hospitals with 123 billion DZD, in support of the Unemployment Insurance Fund of 74 billion DZD and funding for the National Fund for the Equalization of Social Services (FNPOS) (21 billion DZD) (Aziez, 2025).

Regarding health insurance, reimbursements and benefits approached 600 billion DZD, including reimbursement for more than 80 million prescriptions and benefits associated with the health insurance card. These benefits included pharmaceutical expenditures (329 billion DZD), coverage for chronic diseases (230 billion DZD), coverage for childbirth and maternity (5 billion DZD), cardiovascular diseases (6 billion DZD), and renal failure (17 billion DZD) (Aziez, 2025).

Obviously, the previously recorded financial surpluses were temporary as the acceleration of medical expenditures at a rate exceeding the growth of contribution revenues led to a structural deficit through 2022. Although total revenues subsequently improved, the allocation of a significant portion to support other branches, such as the Pension Fund and running of hospitals, would leave the fund with a narrow financial margin to cover its demanding healthcare obligations, which necessitate a review of financing mechanisms and expenditure controls.

6.3.3 Social security (health) card

Chifa (health) Card exempts users from paying upfront for medication or treatment costs up to certain limits, increasing rates of health service utilisation, particularly among specific groups. The widespread use of this card reflects the funds' ability to provide health coverage to various groups (e.g., wage earners, non-wage earners, retirees, and other special groups) across the country. Table 6 presents indicators regarding the distribution and use of Chifa Card.

The table shows that a significant proportion of the population was covered by health insurance, reflecting the

expansion of social protection and significant pressure on insurance fund budgets. This increase requires novel funding sources and strict medical oversight mechanisms to curb excessive or unjustified use of medications and services, ensuring the system's long-term financial sustainability.

Table 6. Chifa Card indicators

Indicator	Value
Number of cards issued	20.5 million cards
Demographic coverage	30 million people (insured persons and dependents)
Share of health expenditure in total social security expenditure	68%
Service digitization	12,000 contracted pharmacies enabling real-time card updates

Within this framework, Algeria's social security system listed 7,400 reimbursable medications; among these, 5,244 were available as brand names reimbursed to patients (Doukha, 2025). This list is updated through joint ministerial decisions issued by the Ministry of Health and the Ministry of Finance, periodically adding new medications or removing others in the Official Gazette.

6.3.4 Health insurance expenditures in Algeria

Based on data collected from the Directorate General of Social Security (DGSS), the main health insurance expenditures in Algeria—focusing on the CNAS—can be outlined as follows:

- **Drug Reimbursements:** Constitute the largest expenditure item in the National Social Insurance Fund for Wage Earners, accounting for 7.45% of expenditures in 1990. By 2025, they had accounted for 87.5% of total health insurance expenditures. With continued rise at an accelerating pace, reimbursements increased from 0.94 billion DZD in 1990 to 190 billion DZD in 2016, while reaching approximately 300 billion DZD in 2025. This increase could be attributed to the rise in the number of reimbursable items.
- **Referrals for Treatment Abroad:** The number of referrals abroad fell dramatically from 6,300 patients in 1985 to 2,100 patients in 1990 due to economic crisis and the decline in hydrocarbon revenues. It then doubled to 4,539 patients in 1994; from 1995 till 2025, the number declined drastically from 2,411 to 72 patients, respectively. Nevertheless, costs associated with this category continued to grow due to the serious and costly nature of illnesses. The figure doubled from 0.76 billion DZD in 1985 to 1.5 billion DZD in 2009, before falling to 0.685 billion DZD in 2025.
- **Flat-rate Contribution for Hospitals:** The flat-rate hospital contribution rose significantly from 120 million DZD in 1973 to 27,021 million DZD in 2004 (a 225.17-fold increase), and in 2005, the flat-rate contribution was set at 35,000 million DZD, rising considerably to 150,000 million DZD by 2025.
- **Daily Benefits:** The Fund's expenditures on daily benefits rose rapidly from 1.34 billion DZD in 1990 to 18 billion DZD in 2015, and in 2019, more than 28.5 billion DZD was lavishly spent due to massive increases in short- and medium-term sick leave as a result of the COVID-19 pandemic. This increase continued through 2025, when it was estimated at 52.5 billion DZD.

7. The Reality of Supplementary Health Insurance in Algeria

Supplementary health insurance in Algeria, whether through social mutual aid societies or commercial and banking insurance offerings, functions as a supporting mechanism designed to settle expenses not covered by the mandatory system. This type of insurance is indispensable. Although it helps finance medical care, it cannot fully cover the continually rising costs of treatment, leaving a financial burden borne either directly by the patient or shifted to supplementary providers (Chaoui & Legros, 2012).

Health insurance functions as a financial safety net to absorb two types of direct costs, i.e., the statutory copayment (20%) for services that are not fully covered, and the price gap resulting from the freeze on reference reimbursement rates since 1987, compared with the actual costs of the private medical sector, particularly specialized surgery, diagnostic tests, and dental care.

The mandatory public system is based on a pay-as-you-go model. Though this mechanism has historically fostered intergenerational solidarity and facilitated access to care, it is under tremendous pressure due to demographic and epidemiological shifts. Life expectancy rose to 77.1 years as of 2023, and the prevalence of chronic diseases affected approximately 25% of the adult population; these trends have been placing pressure on revenue collection, cost management, and the long-term sustainability of the system (Boukhedimi et al., 2025).

Despite the broader growth of the insurance sector, supplementary health insurance in Algeria remains accountable for a marginal share of the total market. Based on data compiled from the CNA, Table 7 details the volume of health insurance premiums generated by life and health insurance companies between 2018 and 2024.

As illustrated from Table 7, the volume of personal insurance policies issued during 2018–2024 rose from 12,629 million DZD in 2018 to more than 16,568 million DZD in 2022, and it reached the peak at 22,530 million DZD in 2024. This type of insurance accounted for only a small percentage of the insurance market in Algeria

compared with property and casualty insurance, which accounted for at least 80%. In contrast, the table demonstrates that health insurance policies accounted for a very small share of personal insurance, reaching approximately 116 million DZD in 2024 at best and occupying a share with less than 1% of the total premium volume for personal insurance policies in Algeria. The weakness of supplemental insurance in Algeria could be attributed to several factors. First, the insurance sector was confined to group insurance policies for large companies, while individual and family underwriting stagnated due to limited disposable income and a weak insurance culture. Second, the exclusion of informal workers from the mandatory system, which automatically deprived them of supplementary and mutual insurance programs linked to formal employment. Third, the coverage was limited to typical accidents and the absence of flexible and innovative insurance products.

Table 7. Share of supplementary health insurance in the premium income of life and health insurance companies during the period of 2018–2024

Year	Total Personal Insurance Premiums (million DZD)	Health Insurance Premiums (million DZD)	Share (%)
2018	12,629	95	0.8
2019	14,118	100	0.7
2020	12,489	50	0.4
2021	13,485	56	0.4
2022	16,568	88	0.5
2023	19,620	106	0.54
2024	22,530	116	0.5

Note: DZD = Algerian dinars.

Supplementary health insurance products in Algeria are offered by personal insurance companies and by several banks as part of their bancassurance offerings. Based on the CNA (2024), Table 8 displays the market share of personal insurance companies in the health insurance sector.

Table 8. Market share of personal insurance companies in the health insurance sector as of 2024

Case No.	Name of the Company	Market Share (%)
1	Cardif El Djazair	89.8%
2	Amana Assurance	9.4%
3	CAARAMA ASSURANCE	0.8%
4	L'Algérienne Vie	0.05%

Note: Percentages may not sum to 100% due to independent rounding.

The table shows that the privately owned foreign-capitalised company Cardif El Djazair held the largest market share in Algeria's supplemental health insurance sector by exceeding 89%. In comparison, the combined share of public companies and their subsidiaries, namely CAARAMA and L'Algérienne Vie, did not exceed 1%. This was mainly due to foreign expertise and banking distribution channels (bank-based insurance), which had strengthened the company's operations in this sector. In contrast, public companies paid less attention to this type of contract.

Insurance companies and mutual insurance societies have the following target groups in their operation of supplemental health insurance:

- **Institutions and companies (Group insurance):** In Algeria, supplementary health insurance can be provided through group insurance arrangements, whereby employers subscribe to health insurance contracts on behalf of their employees. Such arrangements complement the coverage provided by the mandatory social security system by covering healthcare expenses that remain partially or entirely uncovered. (Sekhri & Savedoff, 2006).
- **Individuals and families:** The offerings targeted at individuals and families in the Algerian market with a focus on covering medical costs that exceed the mandatory coverage ceiling. Social security relies heavily on a reference tariff that is often significantly lower than the actual costs charged by private medical clinics. The focus of this supplemental health insurance is therefore on high-cost services which patients bear most out-of-pocket expenses, such as dental care, prescription eyewear, and hospitalisation and surgical costs. (Chaoui & Legros, 2012)
- **Self-employed professionals:** The category of self-employed workers and professionals covered by the CASNOS system represents a strategic segment for supplemental health insurance companies in Algeria. Given the nature of their independent work, they face a high opportunity cost from time-off work due to illnesses. Therefore, supplemental insurance targets them with products that guarantee rapid access to private-sector medical care, ensuring a swift return to economic activity (CNA, 2022).

Claims settled under this branch did not exceed 0.3 million DZD, representing 0.01% of total personal insurance claims in 2023, while claims were estimated at 3 million DZD, representing 0.1% of total personal insurance

payouts in 2024 (CNA, 2024).

8. Challenges Faced by Algeria's Health Insurance System in Its Development

Despite a solid legal and institutional framework, numerous challenges remain, including coverage gaps, inadequate benefits, and limited development of private health insurance, thus revealing structural weaknesses in the national social protection system. The most significant challenges could be highlighted as follows:

- Financial balance of social security funds: Social security in Algeria faces the problem of achieving a financial balance between its expenditures and resources. While costs and expenditures related to covering services and benefits provided are increasing, the value of contributions is declining.
- Outdated compensation code: The code governing the amount of benefits has not yet been revised since its establishment in 1987 by the MTESS, leading to a massive gap between the amount of contributions deducted and the amount of benefits received. Consequently, the predicament discourages individuals from enrolling in social security programs.
- Non-reimbursable medications: There is a very long list of non-reimbursable medications, which forces insured individuals to pay large sums out of their own pockets to purchase these medications, in addition to paying the uncovered portion of reimbursable medications.
- Demographic growth: Demographic growth is among the most crucial challenges affecting the financial stability of social security funds, particularly the increasing number of special categories that contribute at a reduced rate.

To combat the problems found in the health insurance system, a multitude of measures and solutions are imperative, with the most significant ones listed below for consideration:

- Addressing the problem of demographic imbalance and the decline in the base of active contributors: This could be achieved by diversifying sources of revenues through the imposition of targeted health fees, such as indirect taxes on tobacco and sugary products, and by allocating direct subsidies to cover the costs of national solidarity.
- Narrowing the gap between out-of-pocket payments and reimbursement: By establishing a mechanism for periodically adjusting and indexing the approved reference rates for essential medical services, accurate diagnostics, and essential medications.
- Overcoming the stagnation in the supplemental insurance market: By enacting tax incentives and exemptions for individual and small-business insurance policies, and digitally linking mutual aid societies and commercial insurance companies to the Chifa Card system.
- Curbing the rise in pharmaceutical spending and the risk of waste: This would be achieved by accelerating the rollout of standardized electronic medical records and linking them to the health cards for all contracted practitioners and clinics to monitor treatment pathways and combat fraud.
- Alleviating pressure on the public hospital system: This could be achieved by expanding the strategic procurement of services through model contracts with private clinics for specialized surgery and oncology treatment, based on specific contractual rates under a third-party payment system.

9. Conclusions

Algeria's health insurance system has succeeded in expanding demographic coverage and strengthening solidarity through social insurance funds and the Chifa Card system. However, the system has to tackle mounting financial challenges due to demographic changes, the rise in chronic diseases, and outdated reference reimbursement rates, which place great pressure on the contribution-based distribution system. Furthermore, supplemental insurance remains very limited in its role to reduce out-of-pocket medical expenses.

Merely increasing medical spending would not necessarily be an effective approach to ensure the sustainability of Algeria's health insurance system. Other measures to be explored include the adoption of an integrated strategy based on the need to implement innovative financing mechanisms, advance digitization to rationalize consumption, and update reference prices for treatment. The essential role of strengthening institutional integration between mandatory insurance and supplementary insurance offered by mutual aid societies and private insurers is also highlighted, to alleviate the financial burden on the funds.

This study relied primarily on official reports and aggregate data, which preclude direct microeconomic measurement of household behavior and the burden of out-of-pocket health expenditures. Future research could explore the use of household budget surveys to model the impact of direct payments across different social segments and consider the development of actuarial models for supplemental health insurance products tailored to the realities of the national market. The collaborative efforts to successfully modernise the social security sector are determined by the capacity of public policies to balance the efficiency of economic spending with the requirements of social justice, thereby safeguarding the sustainability of the health safety net for the well-being of future generations.

Author Contributions

Conceptualization, K.B., H.A., A.K., and A.D.; methodology, K.B.; software, A.K.; validation, H.A., A.K., and A.D.; formal analysis, K.B. and H.A.; investigation, K.B.; resources, A.D.; data curation, H.A.; writing—original draft preparation, K.B. and A.K.; writing—review and editing, H.A. and A.D.; visualization, A.K.; supervision, K.B.; project administration, K.B. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- Amrani, S. (2012). *Couverture Sociale des Travailleurs Salariés et Non Salariés en Afrique du Nord : Étude Comparative*. Association internationale de la sécurité sociale.
- APRIL International. (n.d.). *Health insurance in Algeria*. <https://www.april-international.com/en/destinations/afrika/health-insurance-in-algeria>
- Arrow, K. J. (1978). Uncertainty and the welfare economics of medical care. In *Uncertainty in Economics* (pp. 345–375). Academic Press. <https://doi.org/10.1016/B978-0-12-214850-7.50028-0>.
- Aziez, B. (2025). *Le DG de la Caisse annonce un solde de +10 % : CNAS, la balance passe au vert*. L'Algérie Aujourd'hui. <https://lalgerieaujourd'hui.dz/le-dg-de-la-caisse-annonce-un-solde-de-10-cnas-la-balance-passe-au-vert/>
- Bloom, D. E., Canning, D., & Sevilla, J. (2004). The effect of health on economic growth: A production function approach. *World Dev.*, 32(1), 1–13. <https://doi.org/10.1016/j.worlddev.2003.07.002>.
- Bougrine, P. (2005). L'assurance maladie de la sécurité sociale : Mise en place des régimes de soins de santé—L'expérience de la Caisse nationale des assurances sociales des travailleurs salariés. In *Association Internationale de la Sécurité Sociale, Conférence Régionale de l'AISS pour l'Afrique*. Lusaka, Zambia. <https://www.issa.int/pdf/lusaka05/1bougrine.pdf>
- Boukhedimi, F. Z., Merad, M. A., & Doucanef Gadiri, F. (2025). Between social protection and complementary insurance: An analysis of health risk coverage in Algeria. *J. Soc. Prot. Res.*, 6(2), 100–121.
- Centre for European & International Social Security Liaison. (2024). *Le régime algérien de sécurité sociale (non-salariés)*. https://www.cleiss.fr/docs/regimes/regime_algerie_nonsalaries.html
- Chaoui, F. & Legros, M. (2012). *Les systèmes de santé en Algérie, au Maroc et en Tunisie : Défis nationaux et enjeux partagés* (No. 13). IPAMED.
- CNA. (2022). *Note de conjoncture au 31 décembre et au 4ème trimestre 2021 du marché national des assurances*. <https://cna.dz/note-de-conjoncture-au-31-decembre-et-au-4eme-trimestre-2021-du-marche-national-des-assurances/21/03/2022/>
- CNA. (2024). *L'assurance algérienne en chiffres 2024*. <https://cna.dz/chiffres/analyse-de-marche/>
- CNAS. (2018). *The presentation of CNAS*. <https://cnas.dz/en/the-presentation-of-cnas/>
- CNESE. (2024). *Développement social et humain : Principales réalisations et enjeux 2019–2023*. <https://www.scribd.com/document/841342965/6-Eme-Session-Pleni%C3%A8re>
- Cutler, D. M. & Zeckhauser, R. J. (2000). The anatomy of health insurance. In *Handbook of Health Economics* (Vol. 1, pp. 563–643). Elsevier. [https://doi.org/10.1016/S1574-0064\(00\)80170-5](https://doi.org/10.1016/S1574-0064(00)80170-5).
- Doukha, Z. (2025). *Projet de loi portant prolongation du congé de maternité : Une prise en charge exemplaire*. El-Moudjahid. <https://www.elmoudjahid.dz/fr/actualite/projet-de-loi-portant-prolongement-du-conge-de-maternite-une-prise-en-charge-exemplaire-236549>
- Eco Times. (2025). *Ses recettes ont augmenté de 10% en 2024 à 182,7 milliards DA : Les voyants de la CNAS sont au vert*. <https://ecotimesdz.com/ses-recettes-ont-augmente-de-10-en-2024-a-1827-milliards-da-les-voyants-de-la-cnas-sont-au-vert/>
- Hamadi, L. & Abrika, B. (2023). La sécurité sociale en Algérie entre transformation et restructuration. *Al-Bashaer Econ. J.*, 9(2), 676–693. <https://doi.org/10.33704/1748-009-002-036>.
- Ho, A. (2015). Health insurance. In *Encyclopedia of Global Bioethics* (pp. 1–9). Springer. https://doi.org/10.1007/978-3-319-05544-2_222-1.
- ISSA. (2024). *Algérie*. <https://www.issa.int/fr/country-profile/algeria>
- Kapur, D. R. (2024). Understanding the meaning and significance of health insurance. *Int. J. Prev. Med. Health.*, 3(5), 24–29. <http://doi.org/10.54105/ijpmh.B1039.03050723>.

- Kutzin, J. (2013). Health financing for universal coverage and health system performance: Concepts and implications for policy. *Bull. World Health Organ.*, 91(8), 602–611. <https://doi.org/10.2471/BLT.12.113985>.
- Manjunatha, N. (2019). Descriptive research. *J. Emerg. Technol. Innov. Res.*, 6(6), 863–867.
- Mabrouk, H. (2019). *Code des Assurances*. Berti Éditions.
- Mossialos, E., Dixon, A., Figueras, J., & Kutzin, J.(2002). *Funding health care: Options for Europe* (No. WHO/EURO: 2002-617-40352-54036). World Health Organization. Regional Office for Europe. <https://iris.who.int/handle/10665/332208>
- MTESS. (n.d.). *National policy and legislation on social security*. <https://www.mtess.gov.dz/fr/politique-nationale-et-legislation-de-la-securite-sociale/>
- ONS. (2017). *Annuaire statistique de l'Algérie*.
- Saltman, R. B., Busse, R., & Figueras, J. (2004). *Social Health Insurance Systems in Western Europe*. Open University Press.
- Sekhri, N. & Savedoff, W. (2006). Regulating private health insurance to serve the public interest: Policy issues for developing countries. *Int. J. Health Plann. Manage.*, 21(4), 357–392. <https://doi.org/10.1002/hpm.857>.
- Sifer, S. & Guehairia, A. (2024). Évolution de la politique de sécurité sociale en Algérie: De la genèse à la crise sanitaire mondiale. *Cah. CEDIMES*, 19(HS), 81–97. <https://doi.org/10.69611/cahiers19-HS-05>.
- Statista. (n.d.). *Health insurance—Algeria*. Statista Market Insights. <https://www.statista.com/outlook/fmo/insurances/non-life-insurances/health-insurance/algeria>
- United Nations Economic & Social Commission for Western Asia. (2025). *Social protection reform bulletin of the Arab region: Issue 1*. https://socialprotection.org/sites/default/files/publications_files/social-protection-reform-bulletin-arab-region-issue-1-english.pdf
- WHO. (2005). *Designing health financing systems to reduce catastrophic health expenditure*. <https://www.who.int/publications/i/item/WHO-EIP-HSF-PB-05.02>
- Ziani, L. (2020). La couverture du risque maladie par l'assurance maladie sociale en Algérie: Etat des lieux et contraintes. *J. Excell. Econ. Manag. Res.*, 4(1), 173–188. <https://doi.org/10.34118/jeemr.v4i1.764>.